

A Lower Bounds of the Cost Function

PROPOSITION 2 (NON-DECREASING LOWER BOUND). For a pack size s , $1 \leq s \leq n$, the non-decreasing lower bound $L(s)$ of the storage cost $C(s)$ is as follows,

$$C(s) \geq L(s) = s \max_{1 \leq i \leq s} y_i + \sum_{i=s+1}^n y_i + \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil.$$

That is, for $1 \leq s_1 \leq s_2 \leq n$, the lower bound $L(s_1)$ must be no larger than $L(s_2)$,

$$L(s_1) \leq L(s_2).$$

PROOF. According to Definition 1, we could get that

$$\begin{aligned} C(s) &= s \sum_{j=1}^{m-1} q_j + (n - (m-1)s)q_m + m \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil \\ &\geq sq_1 + s \sum_{j=2}^{m-1} q_j + (n - (m-1)s)q_m + \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil \\ &\geq sq_1 + \sum_{i=s+1}^{s(m-1)} y_i + \sum_{i=s(m-1)+1}^n y_i + \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil \\ &\geq s \max_{1 \leq i \leq s} y_i + \sum_{i=s+1}^n y_i + \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil = L(s). \end{aligned}$$

For $1 \leq s_1 \leq s_2 \leq n$, we have $\max_{1 \leq i \leq s_1} y_i \leq \max_{1 \leq i \leq s_2} y_i$, and the difference $\Delta L(s)$ between the lower bound $L(s_1)$ and $L(s_2)$ is as follows,

$$\begin{aligned} \Delta L(s) &= L(s_1) - L(s_2) \\ &= s_1 \max_{1 \leq i \leq s_1} y_i + \sum_{i=s_1+1}^n y_i - (s_2 \max_{1 \leq i \leq s_2} y_i + \sum_{i=s_2+1}^n y_i) \\ &= s_1 \max_{1 \leq i \leq s_1} y_i - s_2 \max_{1 \leq i \leq s_2} y_i + \sum_{i=s_1+1}^{s_2} y_i \\ &= s_1 (\max_{1 \leq i \leq s_1} y_i - \max_{1 \leq i \leq s_2} y_i) + (s_1 - s_2) \max_{1 \leq i \leq s_2} y_i + \sum_{i=s_1+1}^{s_2} y_i \\ &\leq s_1 (\max_{1 \leq i \leq s_2} y_i - \max_{1 \leq i \leq s_2} y_i) \\ &\quad + (s_1 - s_2) \max_{1 \leq i \leq s_2} y_i + (s_2 - s_1) \max_{1 \leq i \leq s_2} y_i = 0. \end{aligned}$$

Therefore, we could conclude $L(s_1) \leq L(s_2)$. $\square$

PROPOSITION 3 (NON-INCREASING LOWER BOUND). For a pack size s , $1 \leq s \leq n$, the non-increasing lower bound $L'(s)$ of the cost $C(s)$ is as follows,

$$C(s) \geq L'(s) = \sum_{i=1}^n y_i + \lceil \frac{n}{s} \rceil \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil.$$

That is, for $1 \leq s_1 \leq s_2 \leq n$, the lower bound $L'(s_1)$ must be no smaller than $L'(s_2)$,

$$L'(s_1) \geq L'(s_2).$$

PROOF. According to Definition 1, we could get that

$$\begin{aligned} C(s) &= s \sum_{j=1}^{m-1} q_j + (n - (m-1)s)q_m + m \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil \\ &\geq \sum_{i=1}^{s(m-1)} y_i + \sum_{i=s(m-1)+1}^n y_i + \lceil \frac{n}{s} \rceil \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil \\ &\geq \sum_{i=1}^n y_i + \lceil \frac{n}{s} \rceil \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil = L'(s). \end{aligned}$$

For $1 \leq s_1 \leq s_2 \leq n$, we have $\lceil \frac{n}{s_1} \rceil \geq \lceil \frac{n}{s_2} \rceil$, and the difference $\Delta L'(s)$ between the lower bound $L'(s_1)$ and $L'(s_2)$ is as follows,

$$\begin{aligned} \Delta L'(s) &= L'(s_1) - L'(s_2) \\ &= \sum_{i=1}^n y_i + \lceil \frac{n}{s_1} \rceil \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil \\ &\quad - (\sum_{i=1}^n y_i + \lceil \frac{n}{s_2} \rceil \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil) \\ &= (\lceil \frac{n}{s_1} \rceil - \lceil \frac{n}{s_2} \rceil) \lceil \log(\max_{1 \leq i \leq n} y_i + 1) \rceil \geq 0. \end{aligned}$$

Therefore, we could conclude $L'(s_1) \geq L'(s_2)$. $\square$

B Detailed Description of Datasets

The NEON datasets, sourced from the National Science Foundation's National Ecological Observatory Network, consist of five distinct time-series collections: Air-pressure (AP), Dew point-temp (DT), IR-bio-temp (IR), PM10-dust (PM10), and Wind-speed (WS). These datasets provide near real-time environmental measurements, capturing barometric pressure fluctuations, relative dew point temperatures over aquatic bodies, temperature variations in infrared-sensitive organisms, concentrations of PM10 particulate matter, and wind speed metrics, respectively. The City-temp (CT) dataset archives historical and contemporary temperature records from major urban centers across the globe. Financial data is represented by three stock market indices: Stocks-DE (SDE), Stocks-UK (SUK), and Stocks-USA (SUSA), which contain historical exchange price data for Germany, the United Kingdom, and the United States, respectively. From the InfluxDB repository, we include several specialized datasets: Bird-migration (BM), which tracks avian movement trajectories and positional data; and Blockchain-tr (BTR), which records daily aggregate values of Bitcoin transactions. Geospatial information is encapsulated in the Citylat (CLT) and Citylon (CLN) datasets, containing the latitude and longitude coordinates of cities worldwide. The Food-prices (FP) dataset, curated by the World Food Programme, documents global commodity price records for food items. Cyber-Vehicle (CV) tracks series data pertaining to the operational status of concrete mixer trucks. TY-Fuel (TF) comprises data concerning the fuel consumption of various vehicle engines. TY-Transport (TT) encompasses data associated with the transportation duration of trucks.

C Pruning Rate on Various Datasets

Figure 21 presents pruning rate of our methods with Propositions 1-5 on various datasets. Since Propositions 5 and 4 prunes close to half of the data in most cases, all datasets allow pruning of over

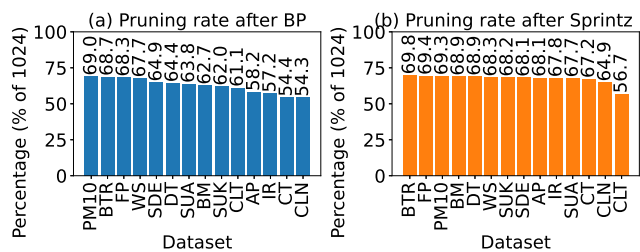


Figure 21: Pruning rate of candidate pack sizes on various datasets

half the candidate pack sizes in Figure 21. Thus, BP-Prune reduces the compression time by more than one third compared to BP-All in Figure 15(c). After processing with Sprintz, the value exhibits greater variability, which more frequently satisfies the preconditions of propositions, thereby enabling even more aggressive pruning than bit-packing. As a result, Sprintz-Prune halves the compression time compared to Sprintz-All, as shown in Figure 15(d).